Solving for angles in right triangles

Core Concepts: Solving for angles in right triangles

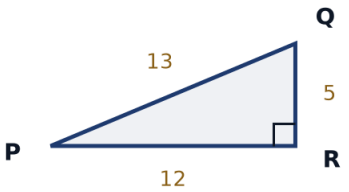
1 In $\triangle ABC$ studying solving for angles in right triangles, $m\angle A = 51^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 84°

b. 94°

c. 76°

d. 129°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm

c. 15 cm

d. 20 cm

3 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 13

c. 11

d. 15

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying solving for angles in right triangles, $m\angle A = 51^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 77°

b. 87°

c. 69°

d. 129°

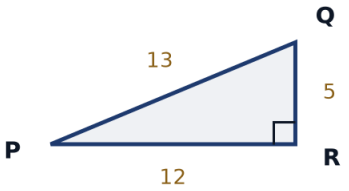
5 A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?

a. 12 cm

b. 15 cm

c. 20 cm

d. 45 cm



6 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$?

a. 11

b. 17

c. 15

d. 19

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying solving for angles in right triangles, $m\angle A = 51^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 70°

b. 80°

c. 62°

d. 129°

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8 A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm

c. 25 cm

d. 80 cm

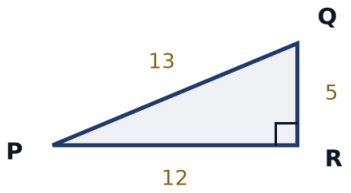
9 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$?

a. 15

b. 21

c. 19

d. 23



Exam-Style Problem Solving & Modeling

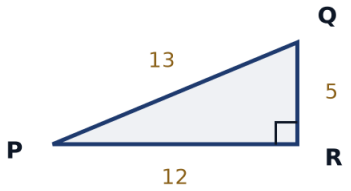
10 In $\triangle ABC$ studying solving for angles in right triangles, $m\angle A = 51^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 63°

b. 73°

c. 55°

d. 129°



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11 A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm

c. 30 cm

d. 125 cm

12 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$?

a. 19

b. 25

c. 23

d. 27

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